

Prepared for: Baltimore Ravens Scouting Department

When looking at standard pressure rate, the number itself lacks context and meaning. Two edge rushers that have the same pressure rate can face different blocking assignments (e.g. Player A experiences a double-team (two blockers assigned to one rusher) at a 15% higher rate than Player B). I built a system to add context to pressure rate numbers and profile how an edge rusher is actually winning their reps.

What I built:

- Created a way to break down where an edge rusher is effective and what type of rush move they are strongest at. This is particularly useful to see if a potential player fits our defensive scheme and what we need at that position, if we wanted to make a trade or scout potential free agent signings.
- Created a way to see an adjusted pressure rate, so we know which players have the highest impact on pressuring the quarterback. This accounts for rushers that are being double-teamed and rewards edge rushers that are able to generate pressure under more difficult blocking assignments.
- Identified which rushers warrant the cost of sending an extra blocker and which rushers are generating significant pressure without opponents committing an extra blocker to stop them.

Player Rankings:

The league average pressure rate is set to 1.0, so a player at 2.0 is twice as effective as the average pass rusher at generating pressure.

Player	Team	Effectiveness	How They Win	Double-Team Rate
Myles Garrett	CLE	2.10	Speed	29%
Za'Darius Smith	MIN	1.83	Speed	41%
Justin Houston	BAL	1.82	Speed	19%
Trey Hendrickson	CIN	1.77	Power	22%
Nick Bosa	SF	1.77	Speed	18%
Micah Parsons	DAL	1.69	Counter	29%

Key Findings:

- **Myles Garrett** sets the standard, operating at 2.1 times the league average effectiveness while being double-teamed on nearly three out of every 10 snaps.
- **Nick Bosa is generating elite pressure without drawing extra attention from opposing offenses. He ranks in the Top 5 in effectiveness at a double team rate of only 18%. This is the most immediate gap that a film study should look to explain.**
- **Za'Darius Smith's greatest value is the pressure he creates for his teammates. Smith gets double teamed on 4 out of every 10 snaps (41%), which opens up many opportunities for his fellow pass rushers to generate pressure on the quarterback by winning their 1-on-1 situations.**
- **Knowing how a pass rusher wins, allows us to prepare a counter.** For example, Trey Hendrickson wins his rushes primarily through power, driving a blocker towards the quarterback to collapse the pocket. This can be disrupted by either using a chip to limit his momentum going into the block, or having a strong enough tackle to anchor down at the point of contact. Each technique profile in this analysis allows us to make an adjustment that can be built into a weekly protection scheme when preparing for matchups.

Limitations:

- This analysis only covers 9 weeks of the 2022 season. These numbers can be influenced by strength of schedule, injuries, and sample size.
- It is likely that a rusher's technique profile reflects the scheme that they play in, and may not be entirely reflective of the player's skill. Moving to a different team or scheme can change the technique profile for an edge rusher, their double-team rate and how effective they are.
- This system is designed to tell you how a rusher wins their reps, but does not explain why certain numbers are the way they are. A low double team-rate is a prime example of this with Nick Bosa. Film is still needed to break this number down even further in order to explain why his double team rate is lower than edge rushers like Myles Garrett and Za'Darius Smith.